The Law of Graphic Equalisation — Physics Grounding

Change, c, the Reconciliation of Special and General Relativity, the Dimensional Constants, and the Second Law

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Abstract

This companion to Graphic Equalisation develops the physics groundings of the change-substrate. Its central move is that a frame's time is an inferred count of *cycles* and that the invariant speed c is the frame's own change-throughput unit, 1 by construction rather than by postulate; from that one economy special and general relativity, the dimensional constants, and the second law follow as *shapes*. Special relativity's c-invariance becomes a theorem; special and general relativity reconcile as one fixed change-budget spent two ways, on motion or on change-load; the dimensional constants $(c, \hbar, k_{\mathrm{B}}, G)$ are each a frame's own base unit and their magnitudes are cross-frame artefacts; the uncertainty principle is the finite-aperture floor, the Landauer bound the information–energy conversion, quantization the aperture cell, and the Planck scale the frontier. The second law is no-copy accumulated, with entropy frame-relative, the demon exorcised by cost, and local order affordable as paid minting. The claims are information-level, physics taken as an embodiment: the paper mechanises the shapes and explicitly does not derive the field equations, the dimensionless constants, or a theory of quantum gravity.

1 Scope

This paper is a grounding, not a physical theory. It inherits Graphic Equalisation's position that the claims are about information — distinction, frame, admissibility, realisation — of which physical theory is one embodiment, and that a physical result bears on an information-level claim when it settles an informational fact. Everything below is at that level. The strong claims are structural: that there is a c, a $\hbar$, an uncertainty principle, a Landauer floor, a second law, and that special and general relativity reconcile as one change-economy. The weak part is anything carrying a number: no field equations, no quantitative metric, no exact Lorentz factor, no dimensionless constant, and no theory of quantum gravity. Each grounding is a *promoted* constraint exposed to falsification, in the manner of Graphic Equalisation's collapse bridge, not a result asserted from within.

2 c as the change-to-time rate

Graphic Equalisation establishes that time is not a substrate dimension but is *inferred* by a frame, whose unit is a *cycle* — one traversal of a closed rule-loop, of no fixed length. Once time is the inferred count of cycles, the invariant speed c has a natural reading, consistent with the missing-denominator result (there is no rate *in* substrate-time, since there is no substrate-time to divide by). c is the fixed *rate* at which change is realised — per unit of the substrate's throughput, a cycle consuming a variable amount of it — from which a frame's time is inferred. So c is prior to time: the conversion between change-count and inferred duration, which is exactly what a constant relating the temporal and spatial parts of an interval is.

Proposition 2.1 (c is each frame's own unit, hence invariant by construction)**.** *c is each frame's own base rate, its change-throughput, so every frame measures its own c as the unit,* 1 *— and a unit cannot be measured as anything else. Its invariance is therefore a theorem, not a postulate: not that all frames happen to agree on a speed, but that each is reporting its own unit.*

A value *other* than 1 is a cross-frame quantity: the familiar number appears only when one frame is compared with another — change-throughput against a spatial metric, seconds against metres — which is where the continuous lives, in the maps between frames rather than in a frame's own discrete states. One can give time a number at all only by comparing frames; in its own frame c is the unit, and its particular value (299 792 458 in SI) is an artefact of the comparison, not a fact about the substrate. And c is a *limit* because a signal is a sequence of changes and none can be realised faster than the frame's throughput.

3 The change-budget: speed, dilation, mass, aperture

Speed is a quantum of that unit. The fixed throughput c splits between *spatial motion* and *internal change* (proper time), and a speed is the fraction spent on motion. At rest the whole budget is internal change and the clock ticks fully; approaching c moves budget to motion, leaving less for internal change, so the clock slows — time dilation as budget re-allocation — and at c the whole budget is motion and no proper time remains. Mass is the budget *reserved* for internal change, so a massive graph can never put all of it into motion: it asymptotes to c and the energy to approach diverges, while a massless one reserves none and *is* at c, with no proper time.

Because observation is funded by internal-change throughput, approaching c coarsens a frame's *aperture*. The transformation between frames in relative motion — time dilation and length contraction — is an aperture transformation between different budget-splits, and each frame's aperture self-calibrates so that its own c still reads 1. The Lorentz factor is the aperture change that holds $c = 1$ across the split. (The constant-magnitude four-velocity of Minkowski geometry (Minkowski, 1909) is this budget-split stated as a metric fact rather than as a mechanism.)

4 Special and general relativity reconciled

The two relativities are one mechanism, because the one fixed budget c can be drained two ways.

Proposition 4.1 (SR and GR are one change-budget spent two ways)**.** *Special relativity is the budget spent on* motion*: a moving graph puts throughput into spatial motion, leaving less for internal change, so its clock slows. General relativity is the budget spent on* change-load*: mass is throughput tied up in internal change, and concentrating change-load (dense mass) consumes the local budget, so local clocks slow. Both dilations are the same act — the one fixed budget drained, by motion or by mass, each leaving less for proper time.*

The equivalence principle is the seam. Accelerating (changing one's motion) and standing in a gravitational field (change-load) draw on the budget identically, so inertial mass (resistance to a change of motion — more state to re-write, hence more change and more cost, the impulse relation of the Narrative-momentum paper with mass as the exchange rate) and gravitational mass (change-load consuming throughput and slowing local time) are one quantity, "how much change this matter loads onto the ruleset," measured two ways. General relativity assumes their equality and does not explain it; here they coincide by construction.

Corollary 4.2 (Relativity reconciles because c is defined, not postulated)**.** *Because c is here* defined *(Proposition 2.1) rather than postulated, special relativity's postulate of c-invariance becomes a theorem, and its effects follow as consequences rather than as separate surprises, with the classical paradoxes dissolving into aperture differences — each moving clock right, each reading its own $c = 1$. Special and general relativity then reconcile (Proposition 4.1) as velocity- and gravity-dilation of one budget, joined at the equivalence principle.*

5 Relativity and collapse

Relativity also reconciles with the collapse picture of quantum measurement, by a division Graphic Equalisation already carries: continuous structure lives in the maps between frames, discrete change in a frame's own states. The continuous geometry of general relativity is a cross-frame *map*; the collapse of quantum measurement is the in-frame *event*; and c — the change-to-time rate — is the seam joining the discrete event to the continuous metric. The two are the map and the event of one change-substrate, not rival ontologies, and the standing incompatibility is the artefact of treating them as rivals. This is a claim about the *shape* of the reconciliation, not a derivation: that the field equations follow is not asserted, and no theory of quantum gravity is offered.

6 The measurement problem is frame-relative collapse

Collapse in Graphic Equalisation is a local write that realises one continuation *relative to the frame that interacts.* It is not a physical process needing a mechanism — no dynamical collapse, no spontaneous localisation — but the in-frame realisation that occurs when frames couple. So there is no frame-independent "when": each frame realises its own outcome upon interaction, and a frame that has not interacted holds no collapse. This is Axiom -1 and relational quantum mechanics (Rovelli, 1996) — facts indexed to interactions, not assigned from nowhere.

Proposition 6.1 (Wigner's friend is the map and the event of two frames)**.** *The friend measures the system and realises an outcome — collapse relative to the friend, who interacted. Wigner, not having interacted with friend-and-system, holds relative to himself the continuous unitary* map *(Section 5). Both are correct: the friend holds the in-frame* event*, Wigner the cross-frame* map*, and these are the two sides of the distinction that reconciles general relativity with collapse. The apparent contradiction requires a single frame-independent fact about whether collapse occurred, which Axiom -1 forbids; without that demand the friend's definite result and Wigner's superposition coexist without strain.*

Corollary 6.2 (Any coupling is a measurement; outcomes are definite; no special observer)**.** *Any coupling is a measurement relative to the coupled frames, so no conscious or privileged observer is required and none is distinguished — a plate, a rock, or a friend measure relative to whatever they couple to. Outcomes are definite because collapse realises* one *continuation, and the basis in which they are definite is set by the distinctions the interaction makes — the aperture of the coupling, which is decoherence selecting the pointer basis. There is no cut and no collapse mechanism to locate.*

This dissolves the measurement *problem* — the when, the how, the who, and the definiteness — but it does *not* derive the Born rule. That outcomes are realised frame-relatively is the shape; that their probabilities go as the squared amplitude is a number, and the framework says nothing about it. The same shape/number division applies: the paradoxes are mechanised, the quantitative probabilities handed back untouched.

7 Entanglement is dimensional access through frames

An entangled pair is one system in the superframe, described by a joint dimension — the correlation — that is a first-class quantity of the superframe and NULL in each subframe individually (Graphic Equalisation's higher-frame first-class structure). Each subframe has dimensional access to it only through its own boundary, and only in part; the full dimension lives one level up, along the structural-depth axis rather than the spatial one.

Proposition 7.1 (A subframe's local state is mixed because the purity is a superframe dimension)**.** *Described alone, an entangled particle has no pure state — it is mixed — because its pure description requires the joint dimension, which is* NULL *in its subframe. Tracing out the partner is not discarding locally-available information; the missing information was never a subframe dimension. The reduced state is mixed exactly because the purity is a superframe dimension, inaccessible from below.*

Corollary 7.2 (Nonlocality is up, not across; Bell forbids only subframe-local variables)**.** *The connection between the parties is not* across *the space between them but* up*, to the superframe they are both subframes of. No cross-frame write passes between the subframes; each accesses the same superframe dimension locally, and the joint fact appears only when their accesses are compared (a further coupling), so no signal is sent. Bell forbids* local *— subframe-level — hidden variables (Bell, 1964), and the framework has none: the correlating variable is a superframe dimension, so the violation is expected, not paradoxical. Kochen–Specker contextuality (Kochen and Specker, 1967) is the access itself — measuring projects the superframe dimension into the subframe's aperture, so a realised value is context-dependent by construction.*

Remark (Spookiness is an aperture artefact: the light-switch identity)*.* The confusion entanglement provokes is identical in structure to that of a person who has never seen a light switch and does not understand electricity: flip a switch, a far light obeys, with no connection inside the aperture, so action at a distance is inferred. In both cases the inference is wrong for the same reason — there is a connection, through a dimension the observer cannot see: the wiring in the walls for the switch (the opaque runner), the superframe's joint dimension for entanglement (structural depth). Spookiness is therefore a property of the aperture, not of the phenomenon; a switch is mundane once one sees the wiring, entanglement once one sees the superframe. This is the sequence science has run before — magnetism was action at a distance until the field, gravity until curvature, disease until germs — each "spooky" a real connection through a not-yet-visible dimension, each dissolved by the aperture widening rather than the phenomenon changing. "We do not understand entanglement" is structurally "I do not understand the light switch."

Honest boundary: seeing the connecting dimension dissolves the *spookiness* — the qualitative action-at-a-distance mystery — not the *numbers*. The Tsirelson bound, the correlation strength, and the Born weights are not derived; they become ordinary physics questions rather than spooky ones, the same shape/number division as elsewhere.

8 The accelerator: forced concurrency and decoherence

The same c bounds concurrency, and this gives the high-energy regime a reading. Within a frame, change is self-change and strictly ordered, so forcing two changes onto one rule-step

is the paid exception; a particle accelerator is the machine that pays it, spending energy to push particles' interactions to the same step, where the frame — built to sequence — cannot order them and the interaction is *decoherent* in that frame. The energy divergence as a particle approaches c is the change-rate ceiling: two changes cannot be crowded closer than a step, and forcing them there costs more the closer one gets. The strong reading is "energy is the price of forced concurrency, and forced concurrency is frame decoherence"; anything with a number is weak.

9 Dimensional constants are frame-units; dimensionless ones are not

The reading of c as a frame's own base unit generalises. Each dimensional constant is the base unit of one of a frame's own economies — 1 in-frame, a number only cross-frame.

- c is the change-rate unit (Section 2).
- $\hbar$ is the *aperture quantum*, the smallest resolvable cell: observation is funded by finite internal-change throughput, so fine resolution in one dimension trades against another and the trade floors at a minimum cell — which is the uncertainty principle, $\Delta x\,\Delta p \geq \hbar/2$ being the finite-aperture budget stated as a bound, with $\hbar$ its quantum.
- k_{B} is the information–energy conversion: Landauer's $k_{\mathrm{B}}T\ln 2$ (Landauer, 1961) — the floor cost of change of Graphic Equalisation's economics — is one bit's change converted from the information frame into the energy frame at that rate.
- G is the change-load–geometry coupling, mass deforming the containing frame.

That physics sets $c = \hbar = k_{\mathrm{B}} = G = 1$ (natural units) is then not a convenience but a recognition: these are frame-units, so they read 1 in a frame's own terms, and their familiar magnitudes are cross-frame artefacts — the frame's unit against a human metre, second, kilogram, kelvin. "Why is c that number" is a question about the comparison, not the substrate.

Proposition 9.1 (The dimensional/dimensionless boundary)**.** *This dissolves the* dimensional *constants and touches the* dimensionless *ones not at all. The fine-structure constant* $\alpha \approx 1/137$ *and the mass ratios cannot be set to* 1 *by any unit choice; they are genuine cross-frame* ratios*, real numbers with no dimension to absorb, and the framework says nothing about their values.*

It sharpens the real question rather than answering it: the numbers that need explaining are the dimensionless ones, while the pseudo-mysteries — why c, why $\hbar$ — are unit conventions. This is the standard observation that the number of fundamental *dimensional* constants is a matter of convention (Duff, 2002), landing on the side that they are units; the dimensionless remainder is left exactly as open as it was found.

10 The Planck scale, quantization, and fine-tuning

Three consequences follow. First, the *Planck scale* is where the dimensional units meet at 1 simultaneously ($\ell_P = \sqrt{\hbar G/c^3}$ and its companions): the aperture cell ($\hbar$), the change-rate ceiling (c), and the change-load–geometry coupling (G) reach their unit at once. Below it a frame cannot resolve, cannot order, and horizons its own geometry, so nothing there is representable — the " ? " of Graphic Equalisation's epistemic boundary with a location. The Planck scale *is* the frontier, which is why physics predicts its own breakdown there: not a failure of theories but the exhaustion of a frame's units, below which the pre-relational substrate begins.

Second, *quantization* is the aperture being cellular: because $\hbar$ is a minimum cell rather than an infinitesimal, action and energy come in $\hbar$-steps, so the "quantum" of quantum mechanics is Graphic Equalisation's discrete sufficiency read at the resolution axis — the granularity of the aperture, not a separate granularity of the world.

Third, the *dimensionless* constants are contingent rule-content: the framework cannot derive α or the mass ratios, but it can say what kind of thing they are — the contingent specifics of this substrate's rule-set (Graphic Equalisation's nothing-compulsory principle), so "why α" is not "why this necessary value" but "why this rule-set," a question answered with contingency, since a different rule-set gives a different α at no extra machinery. This relocates fine-tuning at its root: the inference "the constants are tuned, therefore design or necessity" assumes they are the kind of thing with a necessary value, whereas the dimensional ones are units with no value to tune and the dimensionless ones are contingent, so their being tunable is expected rather than remarkable; and the "why this rule-set" is not merely deferred but answered by self-location (Section 13) — we are the observers this rule-set produces.

11 Symmetry and conservation

A symmetry is a transformation that produces no residual: a change the structure does not feel, free and reversible, a no-op in the container's grammar. Noether's theorem, symmetry $\leftrightarrow$ conserved quantity, is then the cost side of that. A quantity is conserved precisely when altering it would cost — produce residual, require a paid write — so the symmetric direction is free and the quantity perpendicular to it cannot move without paying. What a transformation leaves untouched is what it costs to change.

Proposition 11.1 (Momentum conservation is the free-continuation default)**.** *Space-translation symmetry — the rules the same everywhere — produces no residual under translation, and momentum is conserved. But momentum conservation is already the free-continuation default of Graphic Equalisation's momentum: the un-rewritten Narrative, Newton's first law, continuation costing nothing. Noether's momentum case and inertia are one statement — momentum is conserved because only changing it costs.*

Proposition 11.2 (Energy conservation is contingent on time-translation symmetry)**.** *Time-translation symmetry — the rules the same each cycle — conserves the change-budget, which is energy. The failure case is checkable: where the rules effectively change across cycles there is no time-translation symmetry and energy is not globally conserved. This is the standard general-relativistic result that a time-dependent spacetime has no global energy conservation — in an expanding universe photons redshift their energy away — here from the same principle: rules changing across cycles (cosmic expansion, or minting) means no conserved energy.*

Charge follows from gauge symmetry, which is a redundancy of description: a freedom in the representation that changes no physics, in the framework a cross-frame re-description (in the maps, where the continuous lives) producing no in-frame residual, so charge is conserved.

Remark (The gauge group is contingent rule-content)*.* The framework grounds *that* a symmetry yields a conservation law — the shape, residual-free $\leftrightarrow$ cost-to-change — but *which* symmetries hold, the $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ of the Standard Model, is a fact about this substrate's rule-set, not a necessity. "Why this gauge group" is "why this rule-set," the same relocation as "why α" (Section 10): both are contingent specifics of this universe's grammar, answered with contingency, not derived. The conservation *structure* is universal; the particular symmetries are this universe's.

Honest boundary: this gives why symmetry yields conservation, why momentum conservation is inertia, and why energy conservation fails in cosmology — shapes. It does not derive the Standard Model gauge group, the Noether currents quantitatively, or any number; those are contingent rule-content or ordinary calculations, handed back.

12 The second law

Entropy and its arrow follow from the same economy.

Definition 12.1 (Entropy is what a frame does not distinguish)**.** *The entropy a frame holds is the log-count of the microstructure it carries but cannot tell apart — what falls below its aperture. Entropy is therefore frame- and aperture-relative from the start: a finer aperture resolves more and reads less entropy, a coarser one more.*

This is the Gibbs–Jaynes reading, in which entropy is a property of the observer's coarse-graining rather than of the world in itself (Jaynes, 1957); here it is grounded rather than assumed — entropy is *what this frame does not distinguish.*

Proposition 12.2 (The second law is no-copy accumulated)**.** *Every change discards its predecessor — the prior state goes* NULL *unless paid to keep — so each change pushes structure below the aperture, into the un-distinguished. Change therefore accumulates entropy, because change forgets by default and forgetting is exactly this demotion. The arrow of entropy is the no-copy asymmetry, which is the same asymmetry as accumulated change-cost (the Landauer floors summing) and as causation (a change enables its successor, not its predecessor). Equivalently, entropy is accumulated un-absorbed residual: what no grammar mints stays un-distinguished, so residual piling up is entropy rising.*

Corollary 12.3 (The demon is exorcised, and local order is affordable)**.** *Because k_{B} converts entropy to energy, Landauer's $k_{\mathrm{B}}T\ln 2$ is the price of* resolving *one bit — dragging one distinction up out of the un-distinguished, or erasing one back down. Maxwell's demon is exorcised by exactly this: it must pay to distinguish, and that payment is the entropy it thought it removed for free (Bennett, 1982). There is no free distinction, hence no free entropy reduction. Yet local entropy reduction is possible: minting resolves residual and builds structure, but it is a write, so it is paid, and the payment is Landauer heat exported to the surroundings — local order bought by global entropy increase. This is Schrödinger's "life feeds on negentropy" (Schrödinger, 1944) grounded: an organism, a learner, a growing graph is a region that pays to mint local structure and dumps the cost as heat. Life does not violate the second law; it is the second law with a subsidy.*

Remark (The arrows unify)*.* The thermodynamic arrow (entropy up), the causal arrow (change enables forward), and the psychological arrow (memory is present structure of a discarded past) are not three arrows that happen to align; they are one, the no-copy accumulation of change. Every "why do they agree" puzzle in the arrow-of-time literature is asking why three descriptions of a single asymmetry point the same way, and they do because the asymmetry is one: change does not keep its predecessor.

13 Cosmology: the initial state, the seed, and self-location

By Graphic Equalisation's no-start-end result the event order has no intrinsic first, so the beginning of the universe is not an intrinsic edge but a *chosen entry* — a boundary condition imposed from the containing frame. This is why physics supplies initial conditions separately from its laws: the laws are the ordering, which fixes no start, so the entry must come from outside. The Big Bang is that entry, the seed; what is "before" it is not an earlier time but the frontier, the pre-relational substrate, unrepresentable from within, the same " ? " that sits below the Planck scale (Section 10).

Proposition 13.1 (The Past Hypothesis is a seed, hence a hypothesis)**.** *The low-entropy initial state from which the second law takes its direction — the Past Hypothesis — is a seed: the initial rule-and-state configuration installed from the container. Its long-standing status as an*

assumption rather than a derivation is therefore correct in kind, since a seed cannot be derived from within (a mint cannot authorise its own installation). The arrow of time then splits cleanly: its mechanism *is no-copy accumulation (Section 12), and its* origin *— that there is a low-entropy state to rise from — is the seed. The direction points away from the imposed low-entropy start; the running is no-copy.*

Proposition 13.2 (Why this seed: self-location)**.** *Why* this *seed — and, equally, why this α, this gauge group — is answered, not declined, and the answer is self-location. A map is referenced from "you are here," and the seed is the here: the origin the frame is drawn from, as $c = 1$ is the unit it is measured from. The observer is not independent of it — by frame-relative identity a different seed is a different observer — so "why this seed rather than another" assumes a trans-seed observer who could compare, and there is none (Axiom -1). It is that way because we are this way, and we are this way because it is that way: one determination read from inside, not two facts one explaining the other.*

This inverts the fine-tuning inference. The constants are not tuned *to* a pre-existing observer; the observer is what those constants produce, so the apparent coincidence is the co-determination of observer and seed mistaken for a tuning aimed at a target. It is the same move as being surprised that the Graph Law applies to everything: it applies because that *is* what everything, measured, is — a constitutive identity, not a contingent breadth, the discriminator being independent re-derivation (Graphic Equalisation's composition falsifier). What remains out of reach is only *derivation*: no formula computes α or the seed. The *why* is answered; the *value* is not derived. Reference points take no "why" — they are where the why is asked.

14 Identical particles are functional identity

Graphic Equalisation's Axiom -1 holds that two things with no difference that makes a difference are the same, and its no-frame-free-identity theorem that one cannot say which of two is which independently of an individuating frame. This is the quantum treatment of identical particles exactly: two electrons carry no label, and "which is which" names no difference and so has no content. Classical physics labels identical objects — treats two identical bodies as distinct in principle — and that label is a fiction, a distinction imposed where none exists.

Proposition 14.1 (Quantum statistics, and the Gibbs paradox, are functional identity)**.** *Classical Maxwell–Boltzmann counting labels identical particles and over-counts arrangements; quantum Bose–Einstein and Fermi–Dirac counting treats them as one where they are indistinguishable. The Gibbs paradox — the spurious entropy of mixing two identical gases — resolves by not labelling them, which is Axiom -1, and ties to Section 12: entropy is what a frame distinguishes, so identical particles, having nothing to distinguish, add no mixing entropy. The $1/N!$ Gibbs correction is functional identity made quantitative.*

Corollary 14.2 (Pauli exclusion is "distinct things need a distinction")**.** *If two fermions were in the same state — the same in every quantum number — they would carry no distinguishing distinction, and by Axiom -1 would be one, not two. To be two they must differ somewhere. The exclusion principle is therefore the requirement that distinct things be distinguishable: there is no such thing as two indistinguishables in one state, because that "two" names no difference. Exclusion is the arithmetic of identity, not a force.*

Remark (The honest limit: spin–statistics)*.* Indistinguishability and fermion exclusion are clean here; the boson side is not. Why bosons pile into one state as a count while fermions cap at one, and the minus sign under exchange tied to a 2π rotation, is the spin–statistics theorem, which needs structure this account does not supply. The framework grounds *that* particles are indistinguishable and *why* fermions exclude, and hands the boson/sign/spin–statistics content back untouched — the same shape/number division as elsewhere.

15 Positioning and prior art

The pieces are individually established; the contribution is the conjunction under one change-economy and the mechanisation of shapes prior art postulates, geometrises, or offers heuristically.

Deriving c, and mass as internal motion. That c should be *derived* rather than postulated has precedent by a different route. Ignatowsky (Ignatowsky, 1911) and, cleanly, Lévy-Leblond (Lévy-Leblond, 1976) obtain the Lorentz transformation from the relativity principle together with homogeneity, isotropy, and group structure alone, with an invariant speed *emerging* rather than assumed and its value left empirical; those derivations use symmetry as the engine where the present one uses the change-substrate — c as each frame's own change-throughput unit, 1 by construction. The reading of mass as the budget reserved for internal change has two standing precedents, offered heuristically rather than mechanised: Schrödinger's *zitterbewegung*, the electron's rest energy as internal motion at c (Schrödinger, 1930), developed as a rest-mass-from-internal- circulation interpretation by Hestenes (Hestenes, 1990); and the elementary result that confined massless radiation — light in a box — carries rest mass E/c^2, internalised c-motion appearing as inertia. The budget-split itself is the constant-magnitude four-velocity of Minkowski geometry (Minkowski, 1909).

Relativity as postulate. Einstein's special (Einstein, 1905) and general (Einstein, 1916) relativity take c-invariance and the equivalence principle as postulates; the present account renders the first a theorem (Proposition 2.1) and the second a common cause (Section 4), at the cost of claiming only the shapes.

Entropy and the demon. The observer-relative reading of entropy is Jaynes's (Jaynes, 1957); the resolution of Maxwell's demon by the cost of information is Landauer's and Bennett's (Landauer, 1961; Bennett, 1982); life as locally paid negentropy is Schrödinger's (Schrödinger, 1944). The present account ties all three to one substrate law — cost is the write.

Dimensional constants. That the count of fundamental *dimensional* constants is conventional is Duff's position in the Duff–Okun–Veneziano exchange (Duff, 2002); the present account gives the reason (they are frame-units) and draws the boundary at the dimensionless remainder.

16 Falsifiability commitments

Each is exposed to attack; the scope caveat of Section 1 applies throughout.

1. c **as frame-unit.** Fails if a frame can measure its own base change-rate as anything other than 1, or if c-invariance requires a postulate that cannot be recovered as the unit reading.

2. **One budget, two dilations.** Fails if velocity time dilation and gravitational time dilation cannot be exhibited as the same fixed budget spent on motion and on change-load respectively — if the equivalence principle is not their common cause here.

3. **Dimensional/dimensionless boundary.** Fails if a genuinely dimensionless constant can be set to 1 by unit choice, or if a dimensional constant carries an irreducible value not absorbable into a frame's units.

4. **Second law as no-copy.** Fails if entropy can rise in a frame whose changes preserve their predecessors, or fall in one without a paid write exporting cost.

5. **Indistinguishability is functional identity.** Fails if identical particles carry a frame-free label distinguishing them, or if the Gibbs $1/N!$ correction and Fermi exclusion cannot be read as "distinct things require a distinguishing distinction." (The boson/spin–statistics content is not claimed.)

6. **Measurement is frame-relative collapse.** Fails if a frame-independent fact about whether collapse has occurred is required to describe a Wigner's-friend scenario — if the friend's outcome and the outside frame's superposition cannot both stand as the event and the map of two frames. (The Born-rule probabilities are not claimed.)

7. **Entanglement is a superframe dimension.** Fails if the correlation of an entangled pair is a subframe-local quantity rather than a joint dimension NULL in each subframe, or if a reduced single-particle state can be pure — i.e. if the mixedness is not the inaccessibility of a superframe dimension. (The Tsirelson bound and correlation strengths are not claimed.)

8. **Conservation is the cost side of symmetry.** Fails if a continuous symmetry (a residual-free transformation) carries no conserved quantity, or if a conserved quantity persists across a genuine change in the rules — in particular, if global energy is conserved in a time-dependent (rules-changing) regime such as cosmic expansion. (The gauge group is claimed to be contingent, not derived.)

9. **The initial state is a seed, answered by self-location.** Fails if "why this seed / constant / gauge group" can be posed from a trans-frame standpoint that could have obtained a different one — if there is an observer-independent "we" separable from the seed. (The claim is that the *why* is self-location and the *value* is not derived; both together, not either alone.)

10. **Scope.** The paper claims shapes only; it is falsified as *overclaiming* if any passage is read to derive a field equation, a dimensionless constant, or a quantum-gravity result — none is asserted.

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